QUANTITATIVE METHODS 1

Worksheet 1 – Presenting and organising data

Exercises

Exercise 1 – Population, sample and variable types

A store wants to study all customers who made a purchase in September. It surveys a sample of 80 customers and records:

Variable	Example
Payment method	card, cash, mobile
Number of items purchased	1, 2, 3, ...
Time spent in store	minutes
Satisfaction level	very dissatisfied to very satisfied

1. Identify the target population, the sample and the statistical unit.
2. Classify each variable as nominal qualitative, ordinal qualitative, discrete quantitative or continuous quantitative.
3. Choose an appropriate graph for each variable.
4. For which variables do cumulative frequencies make sense? Explain.
5. Explain why a large sample is not necessarily representative.

Exercise 2 – Discrete distribution

x_i	0	1	2	3	4	5
n_i	3	7	12	10	6	2

1. Check the total count N.
2. Calculate relative frequencies $f_i = n_i/N$ and increasing cumulative frequencies.
3. Draw a stick/bar diagram.
4. Construct the cumulative distribution function $F(x) = P(X \leq x)$.
5. Find the proportion of customers who placed at most three orders.

Exercise 3 – Continuous variable with unequal class widths

Class	$[10, 14[$	$[14, 18[$	$[18, 26[$	$[26, 30[$	$[30, 35[$	$[35, 40[$	$[40, 45[$	$[45, 50[$	$[50, 55[$
n_i	4	12	32	32	60	35	25	5	5

1. For each class, calculate width a_i, relative frequency f_i and count density $d_i = n_i/a_i$.
2. Draw the correct histogram and explain why raw counts cannot be used as heights when class widths differ.
3. Calculate increasing and decreasing cumulative counts and the corresponding cumulative relative frequencies.

4. Draw the cumulative curves.
5. Find the number and proportion willing to pay less than 35 euros, then at least 35 euros.